



IE406 - Machine Learning

Lab-1: Linear Regression using Gradient Descent

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Objective

In this lab, students will:

- Implement Linear Regression using Mini-Batch Gradient Descent.
- Understand the effect of learning rate, batch size and feature normalization on model convergence.
- Evaluate the performance of the learned model on unseen data.
- Visualize the prediction performance of the final model.

Dataset

Use the **California Housing Dataset** available in `sklearn.datasets`.

The dataset should be loaded using:

```
fetch_california_housing()
```

Task 1: Dataset Preparation

1. Load the California Housing dataset.
2. Split the dataset into:
 - Training Set (60%)
 - Validation Set (20%)
 - Test Set (20%)
3. Standardize all input features using only the statistics computed from the training set.

Task 2: Implement Mini-Batch Gradient Descent

Implement Linear Regression using Mini-Batch Gradient Descent.

- Implement the algorithm completely from scratch using only NumPy.
- Do not use any pre-built regression model available in sklearn.
- Implement the Mean Squared Error (MSE) loss function manually.
- The implementation should allow the learning rate and batch size as user inputs.
- Store the training and validation loss after every epoch.

Task 3: Effect of Learning Rate

Train the model using the following learning rates:

10^{-5} , 10^{-4} , 10^{-3} , 10^{-2} , 10^{-1} , 1, 10

Use a batch size of 128.

Plot:

- Epoch vs Training Loss
- Epoch vs Validation Loss

Determine the learning rate that gives the best validation performance.

Task 4: Effect of Batch Size

Using the best learning rate obtained in Task 3, experiment with the following batch sizes:

1, 16, 32, 64, 128, 256, Entire Training Dataset

For each batch size:

- Plot training and validation loss curves.
- Record total training time.
- Record the number of epochs required for convergence.

Select the best batch size based on validation performance.

Task 5: Effect of Feature Standardization

Repeat Task 3 without performing feature standardization.

Compare:

- Training Loss
- Validation Loss
- Training Time
- Number of Epochs

Comment on the importance of feature standardization.

Task 6: Inference on Test Set

Using the best model obtained from the previous tasks:

1. Evaluate the model on the unseen test set.
2. Report the Test Mean Squared Error.
3. Plot Actual vs Predicted values using a scatter plot.
4. Compare the performance on the training, validation and test sets.

Submission Instructions

Upload **only one Jupyter Notebook (.ipynb)** file to Google Classroom.

The notebook should contain all code, outputs and plots.

Name the notebook as:

[studentid]_lab1.ipynb

Any other filename or additional files will not be accepted.